

ATVTI — CCIS Internship Program 2026

Day 2 Study & Build Guide

Morning: HTML Deep Dive | Afternoon: Build the Full Homepage Structure

Day	Day 2 — HTML Deep Dive & Homepage Build
Morning Goal	Master HTML elements: headings, lists, links, images, forms, semantic tags, and accessibility basics
Afternoon Goal	Build the complete homepage structure: hero, three department cards, features row, footer
Prerequisite	Day 1 completed — VS Code, Node.js, Git, GitHub set up; your branch exists with a starter index.html
Deliverable	Fully structured index.html (no CSS yet) pushed to your Day 1 branch or a new day2 branch

PART 1 — MORNING

HTML Deep Dive

1. Headings and Paragraphs

HTML gives you six levels of headings, h1 through h6, used to create a clear content hierarchy — not to make text bigger or smaller. Use them in order, like an outline.

```
<h1>Main Page Title</h1>          <!-- only ONE per page -->
<h2>Section Title</h2>
<h3>Sub-section Title</h3>
<p>A paragraph of regular body text goes inside a < p > tag.</p>
```

Rule of thumb

Each page should have exactly one `<h1>` — it is the single most important heading. Never skip levels for styling reasons (e.g., using `h4` just because it 'looks right'). Screen readers use heading levels to build a navigable outline of your page for blind users.

2. Lists

HTML has two main list types, both built from a wrapper element plus list items.

2.1 Unordered Lists (bullets)

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
```

```
<li>JavaScript</li>
</ul>
```

2.2 Ordered Lists (numbered)

```
<ol>
  <li>Clone the repo</li>
  <li>Create a branch</li>
  <li>Push your code</li>
</ol>
```

Use `<ul>` when order doesn't matter (a list of features). Use `<ol>` when sequence matters (a set of steps).

3. Links

The anchor tag `<a>` creates a clickable link. The `href` attribute holds the destination.

```
<a href="https://atvti.edu.pk">Visit ATVTI</a>
<a href="pages/about.html">About Us</a>      <!-- relative link to another
page -->
<a href="#contact">Jump to Contact</a>        <!-- jumps to an element with
id="contact" -->
<a href="mailto:info@atvti.edu.pk">Email Us</a>
```

- Absolute URL — full address starting with `http://` or `https://`, used for external sites.
- Relative URL — a path relative to the current file, used for pages within your own site.
- `target="_blank"` opens a link in a new tab — use it for external links so visitors don't lose your site.

4. Images

The `<img>` tag embeds a picture. It is a self-closing tag — it has no closing tag and no content between tags.

```

```

alt text is not optional

`alt` describes the image in words. It is read aloud by screen readers for visually impaired users.

If an image fails to load (slow connection, broken path), the `alt` text displays instead.

Search engines also use `alt` text to understand what an image shows.

Write `alt` text that describes the CONTENT, not 'image of...' — e.g. `alt="ATVTI logo"` not `alt="a picture"`.

5. Forms

Forms collect input from users — names, emails, messages, selections. Every form is built from a `<form>` wrapper and a set of input elements.

```

<form>
  <label for="name">Full Name</label>
  <input type="text" id="name" name="name" required>

  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>

  <label for="message">Message</label>
  <textarea id="message" name="message"></textarea>

  <button type="submit">Send</button>
</form>

```

The label-input connection

The `<label for="name">` must match the `<input id="name">` exactly — that 'for'/'id' pair links them.

This connection lets users click the label text to focus the input, and lets screen readers announce what each field is for. A form with inputs but no matching labels fails accessibility standards.

Input Type	Used For
text	Single-line text — names, titles, short answers
email	Email address — browser validates the @ format automatically
password	Hides typed characters with dots
number	Numeric input only, often with up/down arrows
checkbox	Multiple selections from a set of options
radio	Single selection from a set of mutually exclusive options
submit	Triggers form submission (used on a <code><button></code> or <code><input></code>)

6. Semantic Tags

Semantic tags describe the meaning of content, not just its appearance. Before HTML5, developers used generic `<div>` tags for everything. Semantic tags make your code more readable, more accessible, and better understood by search engines.

Tag	Purpose
<header>	Introductory content — usually a logo, title, and/or navigation
<nav>	A block of navigation links (menus, breadcrumbs)
<main>	The primary content of the page — only ONE per page
<section>	A thematic grouping of content, usually with its own heading

<code><article></code>	Self-contained content that could stand alone — a blog post, a card, a news item
<code><footer></code>	Closing content — contact details, copyright, secondary links
<code><aside></code>	Content tangentially related to the main content — sidebars, pull quotes

Why this matters for today's build

Your homepage will use `<header>` for the hero, `<main>` wrapping the page content, `<section>` for the department cards and features row, and `<footer>` for contact info. Each department card is a good candidate for `<article>`, since each one is self-contained content.

7. Accessibility Basics

Accessibility means making your site usable by everyone, including people using screen readers, keyboard-only navigation, or assistive technology. It is not optional polish — it is a core engineering skill.

- alt text on every `<img>` — describes what the image shows, or `alt=""` if the image is purely decorative.
- label for every `<input>` — connected via the `for/id` pair, never relying on placeholder text alone.
- Logical heading order — h1 then h2 then h3, never skipping levels.
- Descriptive link text — "Read more about our Vision School programme" instead of just "Click here".
- Sufficient color contrast — text must be readable against its background (covered more on the CSS day).
- Keyboard navigability — every interactive element should be reachable and usable with the Tab key alone.

Afternoon Build Session

Build the full homepage structure: hero, three department cards, features row, footer with contact info.

PART 2 — AFTERNOON

Build the Full Homepage Structure

8. Today's Build Brief

You are building the complete structural skeleton of the ATVTI homepage. No CSS today — focus entirely on correct, semantic, accessible HTML. The page needs four sections:

1. Hero — a welcoming banner with the institute name and a short tagline.
2. Four Department Cards — Courses, Vision School, Academy, and CCIS, each as a self-contained block.
3. Features Row — a short list of highlights (e.g., expert instructors, hands-on projects, certification).
4. Footer with Contact Info — address, phone, email, and social links.

9. Step-by-Step Build

Step 1 — Continue on Your Branch

Open a terminal inside your project folder and confirm you're on your Day 1 branch (or create a new one for today's work).

```
git checkout yourname-day1
```

Or, if your instructor wants a fresh branch for today:

```
git checkout -b yourname-day2
```

Step 2 — Plan the Structure Before Typing

Sketch the outline as comments first. This is a habit professional developers use — plan the skeleton, then fill it in.

```
<!-- Hero -->
<!-- Department Cards (ATVTI, Vision School, Academy, CCIS) -->
<!-- Features Row -->
<!-- Footer -->
```

Step 3 — Build the Hero

The hero is the first thing visitors see. Keep it inside `<header>`, and use the page's single `<h1>` here.

```
<header>
  <h1>Axiom Technical and Vocational Training Institute</h1>
  <p>Empowering students across Pakistan with practical, industry-ready
  skills.</p>
```

```
</header>
```

Step 4 — Build the Four Department Cards

Wrap the cards in a `<main>` and a `<section>`, with each card as its own `<article>` since each is self-contained content describing a distinct programme.

```
<main>
  <section aria-label="Our Departments">
    <h2>Our Departments</h2>

    <article>
      
      <h3>ATVTI</h3>
      <p>Technical and vocational training in computing, linguistics and
digital skills</p>
      <a href="pages/atvti.html">Learn more about ATVTI</a>
    </article>

  </section>
```

Why `<article>` here and not just `<div>`

Each department card makes sense on its own — if you copy-pasted just that card onto another page, it would still be meaningful. That self-contained quality is the test for `<article>`. A purely cosmetic wrapper with no independent meaning should stay a `<div>`.

Step 5 — Build the Features Row

A short list of highlights, using a `<section>` with an unordered list or repeated small blocks.

```
<section aria-label="Why Choose Us">
  <h2>Why Choose Us</h2>
  <ul>
    <li>Expert instructors with industry experience</li>
    <li>Hands-on, project-based learning</li>
    <li>Recognised certification on completion</li>
    <li>Pakistan-focused curriculum and case studies</li>
  </ul>
</section>
</main>
```

Step 6 — Build the Footer with Contact Info

Close the page with a `<footer>` containing real contact information, structured for both humans and accessibility tools.

```
<footer>
  <h2>Contact Us</h2>
  <address>
    Main Shakriyal Service Road, Near Mubarik Shadi Hall, Rawalpindi, Punjab,
    Pakistan<br>
```

```

    Phone: <a href="tel:+92511234567">+92 51 123 4567</a><br>
    Email: <a href="mailto:info@atvti.edu.pk">info@atvti.edu.pk</a>
</address>
<nav aria-label="Social Media">
  <a href="https://facebook.com/atvti" target="_blank">Facebook</a>
  <a href="https://linkedin.com/company/atvti" target="_blank">LinkedIn</a>
</nav>
<p>&copy; 2026 ATVTI. All rights reserved.</p>
</footer>

```

The <address> tag

<address> is a semantic tag specifically for contact information — physical address, phone, email — for the page's author or organisation. It is commonly placed inside <footer>. Browsers often italicize it by default, but its real value is the semantic meaning it conveys.

Step 7 — Validate Your Accessibility Basics

Before committing, check your file against this list:

- Every has a meaningful alt attribute.
- Every <input> (if you add any forms later) has a connected <label>.
- Only one <h1> exists on the page; <h2> and <h3> are used in logical order.
- Link text describes its destination, not just "click here".
- <main> wraps the primary content and appears only once.

Step 8 — Preview, Commit, and Push

5. Save the file (Ctrl + S) and open with Live Server to check it renders without errors.

6. Stage and commit your changes:

```

git add .
git commit -m "Day 2: Build full homepage structure — hero, department cards,
features, footer"

```

7. Push to GitHub:

```
git push origin yourname-day1
```

(Use your actual branch name — yourname-day2 if you created a new one today.)

10. End-of-Day Checklist

☐	Hero section built inside <header> with a single <h1> and a short tagline.
☐	Four department cards (ATVTI, Vision School, Academy) each built as an <article>.
☐	Every has descriptive, meaningful alt text.
☐	Features row built as a <section> with a clear of highlights.
☐	Footer built with <address>, phone, email, and social links.

☐	Only one <code><h1></code> on the page; headings follow logical order (<code>h1</code> → <code>h2</code> → <code>h3</code>).
☐	Page previewed in Live Server with no visible errors or missing content.
☐	Changes committed with a descriptive message.
☐	Branch pushed to GitHub successfully.

11. Key Concepts Summary

Term	Plain-English Definition
Semantic tag	An HTML tag that describes the meaning of its content, e.g. <code><nav></code> , <code><article></code> , <code><footer></code> .
alt attribute	Text description of an image, used by screen readers and shown if the image fails to load.
label / for / id	The linking mechanism that connects a form label to its input for accessibility.
Accessibility (a11y)	Designing so the site is usable by people with disabilities or assistive technology.
<code><article></code>	A semantic tag for self-contained content that would make sense on its own.
<code><section></code>	A thematic grouping of related content, typically with its own heading.
<code><address></code>	A semantic tag for contact information about a page's author or organisation.
Relative vs absolute URL	Relative points within your own site (<code>pages/about.html</code>); absolute is a full external address.

12. Common Mistakes to Avoid

- Using `<div>` everywhere instead of semantic tags — it works, but loses meaning and accessibility benefits.
- Multiple `<h1>` tags on one page — keep exactly one.
- Empty or missing alt attributes — every meaningful image needs a real description.
- Labels not linked to inputs — always match `for` and `id` exactly.
- Generic link text like "click here" repeated across the page — be specific each time.
- Nesting `<article>` or `<section>` without a heading inside — each should generally start with its own heading.

Semantic, accessible HTML today means far less rework when we add CSS and JavaScript later.